

DUAL-SHANNON ETHERNET

Theory and Practice

DÆDÆLUS Research

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Shannon Channels are traditionally one-way. A transmitter sends out some bits, and a receiver receives those bits—if they already 'know' in which direction and where in the spectrum to look. Channel Capacity (CC), Signal To Noise Ratio (SNR) and reliability are all presented in the context of *positive* probabilities. This document explores a logical extension where information is conserved, and *probabilities can be negative*. We account for these conserved quantities using Common Knowledge (CK) in the link: where both sides “know what the other side knows that they know” the mutual equilibrium state. Information transfer is effected as a departure from equilibrium, by either side of the link initiating a causal sequence of *interactions*. Only when the causal sequence Common Knowledge is *complete* can *both sides* declare (and know) the Information transfer done, and move on to the next. Otherwise these fouled transactions will cause loss and corruption of data structures.

PRINCIPLES & VALUES OF A THEORY

INTRODUCTION

Ethernet needs redesigning from scratch every 50 years. This document explores what we've discovered in the past 50 years, and what might be needed for the next 50.

Some principles and values have stood the test of time, and although we may not have known the deep reasons they were favored by their inventors at the time (Metcalfe et al), we now know how well they work and why we'd like to keep on using them.

Some principles just don't work as claimed, and their inventors should apologize for imposing their misunderstandings on the rest of the world. The End To End (E2E) principle is one of them.

The most damaging aspect of E2E is that it encourages bad behavior of actors in the middle of the network. Because, they claim, they can't solve *every* failure or error, they dump responsibility on the distributed system programmers for *all possible failures*, including the ones they introduce in their quest to sell more of what they've got. They then use that as an excuse, to pour abuse on the endpoints with DReDD. (Dropped, Reordered, Duplicate and Delayed Packets).

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“Most data center networks traditionally operate like the internet, with lossy routers that drop packets when queues are full. RDMA requires a lossless semantics cost efficiency that is often equated to utilization. But this ignores the externality of redundant traffic and fouled transactions.” [3]

Fouled transactions aren’t the only externality imposed on the infrastructure owners and users. It wastes CapEx, OpEx, HuEx (Human Attention) and EnEx (Energy Expense), dumping wasted bandwidth and retried computations into our ecosystem as CO₂.

SHORT RANGE

DÆ (Distributed Atomic Ethernet) is designed to work only in (local) short range environments, such as within a rack, between modules on a Motherboard or within a Module connecting multiple Chiplets. Consequently, much of the cost and complexity of designing for long range cables is eliminated. No cables are longer than 0.5 m, and it works all the way down to the smallest connections in UCle.

By being local only, we can make very different assumptions about the reliability, failure modes and persistence characteristics of connections. Because our scouting and routing protocols are dynamic and guarantee atomic failover from one route to another without skipping a beat, or losing information. For example, we can even take connections out of service temporarily, re-train them and make them available as a resource pool to the provisioning service. Only we do it with promises rather than imposition.

But wait, we’re getting ahead of ourselves. This document is intended to describe what new principles we might use to build a new Ethernet that we would want to take to Mars, where hopes of resilience and reliability, and a dearth of on-site technicians and SRE’s threatens our daily survivability.

Part One of this document describes what’s new. For those who have a low attention span and want to know why it’s a big deal. Because Part One will be dismissed by the vast majority of intelligent fools and naysayers who find it challenging to think outside their particular box that forwards their intellectual or financial interests, we have three more parts:

Part Two describes the details of the story. How the knowledge learned from physics can be applied to solve what appear to be otherwise intractable problems in distributed systems. This provides the justification to invest time and energy (and money) in the development.

Part Three shows how a Bipartite Shannon Ethernet can be applied to the Analysis of Two-Phase-Commit (2PC): the canonical transaction protocol used in all distributed systems. We then show how it works for 1PC, 3PC 4PC, 5PC, 6PC, 7PC and 8PC with a simple *reversible* protocol built into the Ethernet Link.

Part Four provides a FAQ - for every question we have heard from naysayers (including ourselves). It may provide detailed calculations and comparisons, it will often provide references to the literature, and it will occasionally blow your mind—if you are willing for it to be blown. If not, then we don’t wish to waste your time and wish you well, or as the Germans would say [Guten Tag](#).

Not included in this document, is how this link protocol can be made *compositional* with the Scouting and Routing to build and tear down graph structures in the environment. In our hypothetical unmanned Mars installation, these are the self-configuring, self-healing and self-optimizing protocols we hope will be present so that we don’t have to manually babysit (monitor and respond) to network and other failures in the infrastructure your life may depend upon.

Self-configuration is not only useful on Mars; it's useful on Earth too, especially where the complexity of manual configuration and response to alarms (false or not) makes slaves out of human administrators. Then we swamp those human beings with far more than they can cope with, and the exponentiating complexity needs new levels of abstractions those human "archestrators" can once again be king of.

OVERVIEW OF INFORMATION THEORY

Like mass and energy, Information can be measured, calculated and exchanged. Shannon proved that information can be communicated, with vanishingly small error, at a rate which is limited only by the Channel Capacity.

Bits are not Binary Digits. Information is surprisal, and it's measured in bits. If we use logarithms to base 2, then Shannon information or surprisal of *each outcome* is measured in bits (per usage, not per second).

$$\text{Shannon Information} = \log \frac{1}{p(x_h)} \text{ bits,}$$

And because of the way logarithms work,

$$= -\log p(x_h) \text{ bits.}$$

ENTROPY IS AVERAGE SHANNON INFORMATION

Entropy is a measure of uncertainty, defined as average Shannon Information.

Entropy and information are two sides of the same coin, whether we call a given quantity information or entropy depends on whether it is being given to us or taken away.

Thus receiving an amount of information is equivalent to having exactly the same amount of entropy (uncertainty) taken away.

Mutual information eliminates the technical difficulties associated with the entropy of continuous variables because it involves the difference between two entropies.

Gaussians are important because no other distribution can provide as much information at lower energy cost per bit. Along that line we relate everything to (the origin problem in Cartesian coordinates).

One challenge with Gaussians is the arbitrary choice of the beginning and end of the line (domain), and "where" the origin (zero) is. This reflects closely with the 'arrival time' problem in Quantum Theory.

A Gaussian Channel has an arbitrary labelled origin (0). While the sum of two independent Gaussian channels is also a Gaussian, this is true only if they share the same timeline, and origin. This will become important as we discuss Indefinite Causal Order (ICO), which would violate this assumption about the sum of two independent Gaussian Channels.

MAXIMUM INFORMATION CAPACITY.

Every Datum conveys as much information as possible.

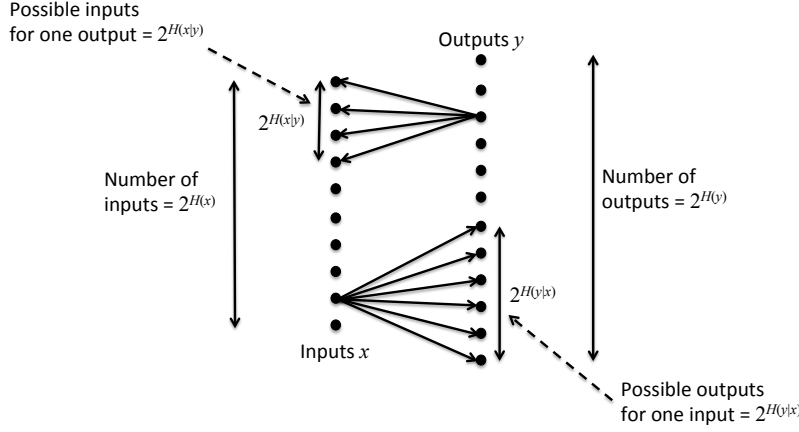


Figure 1: [ADD FIGURE FOR Fan Diagram — extend to PAM/Phase Shift]. Number of inputs that can map to outputs, and number of outputs that can map to inputs, with Qubits.

NOISE REDUCES CHANNEL CAPACITY (DISCRETE, EQUIPROBABLE EXAMPLE).

If the number of equiprobable input states is m_x , then $H(x) = \log_2 m_x$. If there are m_η equiprobable noise values, then $H(\eta) = \log_2 m_\eta$. The output then has $m_y = m_x m_\eta$ equiprobable states, so

$$H(y) = \log_2 m_y = \log_2 m_x + \log_2 m_\eta = H(x) + H(\eta).$$

Thus, the mutual information is

$$I(x; y) = H(y) - H(y|x) = H(x) - H(x|y).$$

FAN DIAGRAM COUNTS (CONCEPTUAL).

$$\text{Number of outputs} = 2^{H(Y)}$$

$$\text{Number of inputs} = 2^{H(X)}$$

$$\text{Possible outputs for one input} = 2^{H(Y|X)}$$

$$\text{Possible inputs for one output} = 2^{H(X|Y)}.$$

CHANNEL REVERSIBILITY

“An important perspective: In exploring the nature of a channel in terms of channel noise, we can pretend to reverse the direction of the data along the channel. Accordingly, before we ‘receive’ an input value we can know that the output can be one of say 16 values.”

“Somewhat counter-intuitively, the average amount of information gained about the output when an input value is received is the same as the average amount of information gained about the input when an output value is received, $I(x; y) = I(y; x)$. This is why it did not matter when we pretended to reverse the direction of data through the channel.” [2]

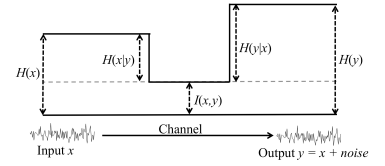


Figure 2: [ADD FIGURE For Mutual Information Box. Use Symboly to show information is carried in 5 ball Newtons Cradle.]

This "Complimentary" view of the 'time direction' of data movement, can half jokingly be referred to as time going backwards, but that just takes everyone down the slippery slope of time machines and killing your grandfather. Instead, we think in a much more (local) quantum mechanical framework, and engineer "logical reversibility" in an Ethernet Link. This is described in detail in the document DAE Bits-V4.pages.

For a rigorous theoretical perspective on this, see Lucien Hardy [Z].

These are the bits we might view on a 100 GHz (PAM4) Ethernet symbol. But we can reduce it by "knowing" that only some of them are possible.

In a Dual Shannon Channel Ethernet, we "know" only certain sequences are possible (because both sides of the link are trying to *comply* with the same 'standards specification' (this is another Shannon Analysis - interoperability)).

In this specification, we require the sender to keep a copy of the last set of bits they sent (EPI - out), and for the receiver to keep a copy of the last set of bits they received (EPI- in). The product of those two represents the ONT (Ontological State) of the transfer.

As an example, for 4 bits on both sides, the product $4 \times 4 = 16$. And the power set of all 16 bits represents all possible combinations is $2^{16} = 65,535$; only a tiny subset of which can be correct (relative to the current "session", depending on what "question" is being asked on the Shannon Channel.

If the question is "Is Anybody There?" then the answers can be, Yes, No, Maybe, and Unknown (nothing heard yet). These can be easily encoded into 01, 10, 11, and 00. This is the basis of an introduction (Hello) protocol. There are many "meanings" that can be attributed to these states, but as we are reminded by the mathematics of Combinatorics [Wolfram], it's all rewriting rules.

Instead of just rewriting rules a la Stephen's Cellular Automata, we use tools from Interaction nets.

Back to Information Theory Overview.

PART ONE: SHANNON CHANNELS

ONE WAY SHANNON CHANNEL

A One Way Shannon Channel has *one side transmitting* and *the other side receiving*. There is no notion of Existence $\exists$; Transmitters can blithely transmit their Information into the universe with no expectation, or Knowledge, that even one receiver exists.

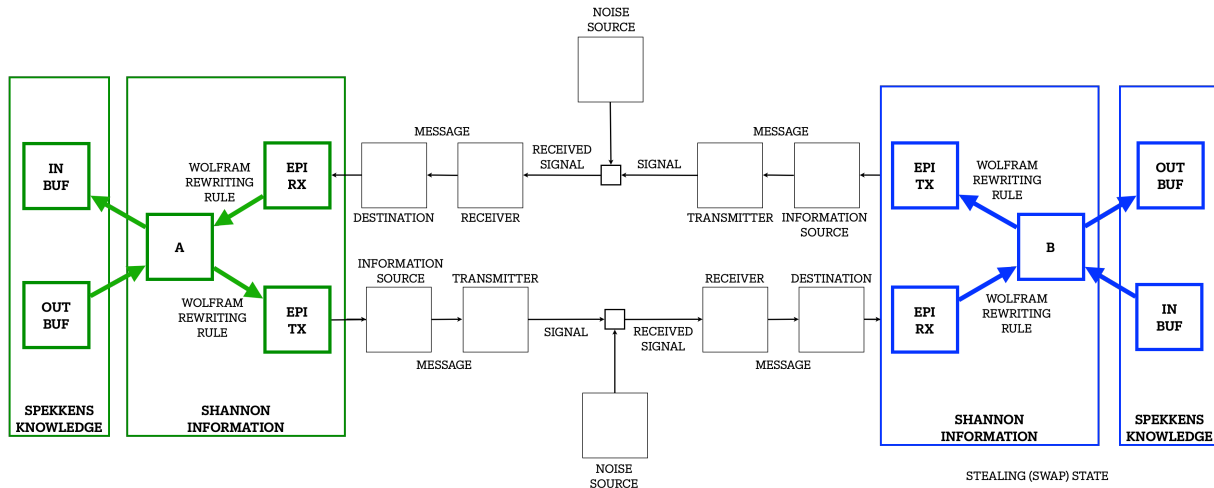
From our Göd's Eye View we look down on the pair of devices which make up the Shannon Channel. Probability is used to mathematically formalize and characterize the one way link. We have already committed the original sin, thinking we know something we don't.

Current Ethernet Links (and all networking built on it) are treated as **two independent** Shannon Channels. One where Alice's Transmitter is connected to Bob's Receiver, and a second one, where Bob's Transmitter is connected to Alice's Receiver. Networking today is designed to sell bandwidth, so they stick steadfastly to a principle of independently "flowing" streams of bytes through either channel in one direction or another.

SHANNON INTERACTION MACHINE

TWO WAY SHANNON CHANNEL

In the language of Shannon’s information theory, a single (one-way) channel focuses on *correlation* [X] and *dissemination* [Y], measuring how well information is transmitted and received. In contrast, a two-way (back to back) channel leverages bidirectional communication to monitor and maintain link health, providing mechanisms for error detection, correction, and performance measurement. Understanding these distinctions enhances our ability to design and manage effective communication networks, ensuring reliability and efficiency in information transmission.



The physics behind this model are based on the Toy Models of Spekkens and Hardy. The key insight is what we can capture on a single clock of an FPGA - an entire 8-byte slice, which represents “simultaneous change”. Every tick of the clock (let’s say 1 ns) is asking a question of the other party. If we got what we were expecting (e.g. liveness), that is not information, and the slice is discarded (after rewriting and returning to the sender).

ERROR MANAGEMENT PHILOSOPHY

A One Way Shannon Channels needs to provide enough “one way redundancy” to maximize the channel capacity [1,2].

A Two Way Channel (back-to-back) ...

BIPARTITE SHANNON ETHERNET

INTERACTION NETS ON AN ETHERNET LINK

An Analysis of the Two-Phase Commit Protocol in the Language of Claude Shannon’s Mathematical Theory of Information.

In a Dual Shannon Channel Ethernet, we “know” only certain sequences are possible (because both sides of the

link are trying to comply with the same ‘standards specification’—this is another Shannon analysis: interoperability).

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ANALYSIS OF TWO-PHASE COMMIT

INTRODUCTION

The two-phase commit (2PC) protocol is a fundamental algorithm in distributed systems to ensure atomic transactions across multiple nodes. Claude Shannon’s *Mathematical Theory of Information*, seminal in the field of information theory, provides a framework to analyze the reliability and efficiency of such protocols. This essay explores the 2PC protocol through the lens of Shannon’s concepts of entropy, information content, and communication channels.

OVERVIEW OF TWO-PHASE COMMIT

The 2PC protocol is designed to achieve consensus in a distributed system where nodes must either all commit to a transaction or all abort, ensuring consistency. It involves two main phases:

1. **Preparation Phase:** The coordinator node sends a "prepare" message to all participant nodes and waits for their responses.
2. **Commit Phase:** Based on the responses, the coordinator sends a "commit" or "abort" message to all participants.

This protocol ensures that either all nodes commit the transaction or none do, maintaining atomicity and consistency across the system.

SHANNON’S INFORMATION THEORY CONCEPTS

Claude Shannon’s theory introduces several key concepts that are relevant to understanding and analyzing the 2PC protocol:

Entropy: Measures the uncertainty or randomness in a set of messages.

Mutual information: Quantifies the amount of information obtained about one random variable through another.

Channel Capacity: The maximum rate at which information can be reliably transmitted over a communication channel.

INFORMATION CONTENT AND ENTROPY IN 2PC

In the context of 2PC, the messages exchanged between the coordinator and participants can be analyzed in terms of their information content and entropy. Each message (prepare, commit, abort) has an associated information content, denoted by $I(x) = -\log P(x)$, where $P(x)$ is the probability of the message x occurring.

1. *Preparation Phase*: The coordinator sends a "prepare" message to all participants. The entropy H of this message is given by:

$$H(\text{prepare}) = -\Pr(\text{prepare}) \log_2 \Pr(\text{prepare}).$$

2. Since this is a deterministic step, the entropy is minimal, reflecting low uncertainty.
3. *Response from Participants*: Each participant's response (yes or no) has its own entropy. If $P(\text{yes})$ and $P(\text{no})$ are the probabilities of a positive and negative response, respectively, the entropy is:

$$H(\text{response}) = -\Pr(\text{yes}) \log_2 \Pr(\text{yes}) - \Pr(\text{no}) \log_2 \Pr(\text{no}).$$

This reflects the uncertainty in the participants' readiness to commit.

MUTUAL INFORMATION AND COORDINATION

The mutual information $I(X; Y)$ between the coordinator's decision and the participants' responses quantifies the reduction in uncertainty about the coordinator's decision after observing the participants' responses. It is defined as:

$$I(X; Y) = H(X) - H(X|Y)$$

where $H(X)$ is the entropy of the coordinator's decision (commit or abort) and $H(X|Y)$ is the conditional entropy given the participants' responses.

COMMUNICATION CHANNELS AND RELIABILITY

The communication channels between the coordinator and participants are critical for the reliability of the 2PC protocol. According to Shannon's channel capacity theorem, the capacity C of a channel is the maximum rate at which information can be reliably transmitted:

$$C = \max_{P(X)} I(X; Y).$$

For the 2PC protocol, ensuring that the communication channels have sufficient capacity to handle the volume of messages within the time constraints is crucial. Any loss or delay in messages can lead to uncertainty and potential inconsistencies.

ERROR DETECTION AND CORRECTION

In practical implementations, error detection and correction mechanisms are essential to mitigate the impact of unreliable communication channels. Shannon's theory provides the foundation for designing such mechanisms. The use of checksums, acknowledgments, and retransmissions helps ensure that messages are correctly received, thus reducing the entropy associated with communication errors.

CONCLUSION

Analyzing the two-phase commit protocol through the lens of Shannon's Information theory provides valuable insights into its reliability and efficiency.

The concepts of entropy, mutual information, and channel capacity highlight the importance of minimizing uncertainty and ensuring robust communication channels. By applying Shannon's principles, we can better understand the challenges and design more effective protocols for distributed systems, ensuring consistency and reliability in transaction processing.

EXPLORING SINGLE AND TWO-WAY SHANNON CHANNELS: CORRELATION, DISSEMINATION, AND LINK HEALTH STATUS MONITORING

INTRODUCTION

Claude Shannon's theory of communication channels is foundational in understanding information transmission in networks. A single (one-way) Shannon channel is often associated with concepts like correlation and dissemination, while a two-way (back to back) Shannon channel can inherently monitor its own link health status. This essay delves into these concepts, exploring how they function and their implications for communication networks.

SINGLE (ONE-WAY) SHANNON CHANNEL: CORRELATION AND DISSEMINATION

A one-way Shannon channel transmits information from a sender to a receiver. In this context, we explore two key aspects: correlation and dissemination.

CORRELATION

Correlation in the context of a one-way channel refers to the degree to which the transmitted and received messages are related. Shannon's information theory quantifies this through mutual information $I(X; Y)$, which measures how much information the received message Y reveals about the transmitted message X .

1. *Entropy and Mutual Information:* For a message X with entropy and a received message Y with entropy $H(Y)$, the mutual $H(X)$ information is:

$$I(X; Y) = H(X) - H(X|Y)$$

This represents the reduction in uncertainty about given X

2. *Correlation Coefficient:* In some cases, the correlation coefficient ρ can be used to quantify the linear relationship between X and Y . For continuous variables, it is given by:

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

where $\text{Cov}(X, Y)$ is the covariance, and σ_X and σ_Y are the standard deviations of X and Y .

DISSEMINATION

Dissemination refers to the process of spreading information from the sender to one or multiple receivers. The effectiveness of dissemination can be measured by how well the information is preserved and interpreted by the receivers.

1. *Channel Capacity*: The channel capacity C is the maximum rate at which information can be reliably transmitted. For a one-way channel, it is defined as:

$$C = \max_{P(X)} I(X; Y).$$

2. This capacity determines the potential for dissemination, ensuring that the transmitted information is accurately received and interpreted.
3. *Broadcasting and Multicasting*: In a dissemination context, broadcasting (sending information to all nodes) and multicasting (sending information to a specific group) are common. The efficiency of these processes relies on the channel's ability to handle multiple receivers and maintain the integrity of the information.

TWO-WAY (BACK TO BACK) SHANNON CHANNEL: LINK HEALTH STATUS MONITORING

A two-way Shannon channel involves bidirectional communication, where messages are exchanged between two parties. This setup allows for inherent link health status monitoring through various mechanisms.

FEEDBACK AND ACKNOWLEDGMENT

In a two-way channel, feedback from the receiver to the sender plays a crucial role in monitoring the link's health.

1. *Acknowledgments (ACKs)*: The receiver sends acknowledgments to the sender upon successfully receiving a message. The absence or delay of ACKs indicates potential issues in the link.
2. *Negative Acknowledgments (NAKs)*: When errors are detected, NAKs inform the sender that retransmission is required. This feedback helps in identifying and correcting errors, maintaining the link's reliability.

ROUND-TRIP TIME (RTT)

Round-trip time (RTT) is the time taken for a message to travel to the receiver and for the acknowledgment to return to the sender.

1. *RTT Monitoring*: By continuously measuring RTT, the system can detect delays or disruptions in the link. Increased RTT may signal congestion, while consistent RTT values indicate a healthy link.
2. *Jitter*: Variation in RTT, known as jitter, can also be monitored. High jitter values may indicate instability in the communication channel, prompting further investigation and possible corrective actions.

ERROR DETECTION AND CORRECTION

Two-way channels facilitate advanced error detection and correction mechanisms, enhancing link health monitoring.

1. *Error Correction Codes (ECC)*: Techniques like forward error correction (FEC) allow the receiver to detect and correct errors without needing retransmission, improving reliability.
2. *Retransmission Protocols*: Protocols such as Automatic Repeat reQuest (ARQ) rely on acknowledgments and timeouts to ensure message integrity. If a message is not acknowledged within a specific time, it is retransmitted.

COMPARATIVE ANALYSIS: ONE-WAY VS. TWO-WAY CHANNELS

1. *One-Way Channel*: Primarily used for dissemination, its effectiveness is measured by correlation (mutual information) and the channel capacity. However, it lacks inherent mechanisms for immediate error detection and correction.
2. *Two-Way Channel*: Provides built-in feedback for error detection, correction, and link health monitoring through acknowledgments, RTT measurements, and jitter analysis. This makes two-way channels more robust for maintaining reliable communication.

4: FREQUENTLY ASKED QUESTIONS

LITERATURE REVIEW

A significant body of literature extends Shannon's information theory into the realm of distributed systems and the two-phase commit (2PC) protocol. Contributions include:

1. **Error Detection and Correction**: Modern research often explores error correction within distributed systems using concepts from Shannon's information theory. For instance, Mukherjee's work on architectural design for soft errors delves into hardware and software techniques to improve reliability, emphasizing error detection and correction mechanisms in distributed environments. This is particularly relevant when considering the reliability of the 2PC protocol, where communication errors can lead to inconsistencies in transaction states (Mukherjee, 2017). [4]
2. **Information-Theoretic Approaches to Feedback****: Elia's research on control-oriented feedback communication schemes examines how feedback can be used to maintain the health of communication links, aligning with Shannon's principles of information theory. This approach can be directly applied to the 2PC protocol's reliance on acknowledgments and negative acknowledgments to ensure that all nodes in a distributed system are synchronized (Elia, 2004). [5]
3. **Causal Relations and Entropy**: Recent advancements in understanding causal relations and entropy in information systems provide a deeper insight into how distributed protocols like 2PC can be optimized. For example, studies presented at the IJCAI conference have focused on leveraging information-theoretic measures to

better understand and optimize the generalization and reliability of distributed algorithms (Dong et al., 2023). [6]

4. 4. Caching and Data Dissemination: Research on caching systems, such as the work by Yu et al., explores the rate-memory tradeoff in caching networks. These principles can be extended to understand how data dissemination in one-way channels can be optimized, ensuring that critical transactional data in 2PC is effectively propagated to all necessary nodes (Yu et al., 2017). [7]

These studies illustrate how Shannon’s theoretical framework can be applied to modern distributed systems, providing tools to enhance the reliability and efficiency of protocols like 2PC. By integrating concepts such as error correction, feedback mechanisms, and entropy analysis, researchers can develop more robust distributed systems that maintain consistency and reliability in the face of communication challenges.

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